

Day 5 — Mixed Cases: One Above, One Below

Module: Nikhila Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit multiplication where one factor is *above* 100 and one is *below*, handling the negative right-part correctly.

Materials

- Whiteboard.
- Drill sheet (8 mixed-sign problems).
- `activities/activity-01-mixed-signs.md` worksheet.

Lesson flow

Warm-up (5 min)

- Complement-to-100 round.
- Excess-from-100 round.
- Now mix: teacher says “97,” class says “–3.” Teacher says “103,” class says “+3.” This trains the sign attention.

Core (15 min) — One above, one below

1. **Set up 102×98 .** Write on board:

$$\begin{array}{r|l} 102 & +2 \\ \times 98 & -2 \\ \hline ? & ? \end{array} \quad \begin{array}{l} \leftarrow \text{excess} \\ \leftarrow \text{deficit} \end{array}$$

2. **The left part — either cross still works:**

- $102 + (-2) = 102 - 2 = 100$.
- $98 + 2 = 100$. (Same answer ✓.)

3. **The right part — sign matters:**

- $(+2) \times (-2) = -4$.
- The right part is **negative**!

4. **What does it mean to have a “negative” right part?**

- Concatenating “100” and “–4” gives... $100 \mid -4 = 100 \times 100 + (-4) = 10000 - 4 = 9996$.
- Check: $102 \times 98 = 9996$. ✓

5. **The “borrow” maneuver — cleaner mental version.** Same problem:

- The negative right part of -4 says: “subtract 4 from the natural concatenation $100|00 = 10000$.”
 - So $102 \times 98 = 10000 - 4 = \mathbf{9996}$. Clean.
6. **Or — the “borrow from left” form:** if students prefer always-positive right parts:
- Left = 100. Right = -4 . Borrow 1 from left (which is worth 100 in the right):
 - New left = $100 - 1 = 99$. New right = $100 - 4 = 96$ (now positive).
 - Answer: **9996**.
7. **Worked examples in chorus:**
- 104×96 : excess +4, deficit -4 . Left = 100 (either cross). Right = $+4 \times -4 = -16$. Answer: $100 \times 100 - 16 = \mathbf{9984}$. ✓
 - 97×103 : deficit -3 , excess +3. Left = 100. Right = $-3 \times +3 = -9$. Answer: **9991**. ✓
 - 108×95 : excess +8, deficit -5 . Left = $108 - 5 = 103$ (or $95 + 8 = 103$ ✓). Right = $+8 \times -5 = -40$. Answer: $103 \times 100 - 40 = 10300 - 40 = \mathbf{10260}$. ✓

8. **The pattern (write on board):**

Case	Cross	Right part sign
Both below	Subtract	+ (deficit $\times$ deficit)
Both above	Add	+ (excess $\times$ excess)
Mixed	Either form — still cancels	– (excess $\times$ deficit)

Activity (20 min) — [activities/activity-01-mixed-signs.md](#)

Brief: - 8 problems on the activity sheet, all mixed-sign. - 8 min individual silent. - 7 min paired check. - 5 min: students re-do any they got wrong, OR move to challenge problems if all correct.

Wrap-up (5 min)

- Quick chorus check: $102 \times 98?$ (9996.) $104 \times 96?$ (9984.) $108 \times 95?$ (10260.)
- Preview Day 6: “Tomorrow — base 1000. The pad becomes 3 digits.”

Student-facing content

Mixed case: one above, one below

1. Find the **excess** of the bigger number and the **deficit** of the smaller (or just signed: + for above 100, – for below).
2. LEFT: cross-subtract using signs. $102 + (-2) = 100$. Either cross still works.
3. RIGHT: multiply the two signed numbers. The RIGHT part is **negative**.
4. Subtract the absolute value of the right from LEFT $\times 100$.

Examples: - $102 \times 98 \rightarrow 100 | -4 \rightarrow 10000 - 4 = \mathbf{9996}$ - $104 \times 96 \rightarrow 100 | -16 \rightarrow 10000 - 16 = \mathbf{9984}$ - $108 \times 95 \rightarrow 103 | -40 \rightarrow 10300 - 40 = \mathbf{10260}$

Tip: for $(a+x)(a-y)$ with $a = \text{base}$, the left is $a + x - y$ and the right is $-xy$.
Whenever excess equals deficit, the left is exactly the base and the answer is $\text{base}^2 - xy$.

Homework

- 10 mixed problems (mix of both-below, both-above, and mixed-sign). Decide the case first.
- Find ONE example in your textbook where mixed-sign Nikhilam would be faster than long multiplication. Bring it.

Differentiation

- **Tier up:** Try 121×79 (further from 100). Excess +21, deficit -21. Cross: $121 - 21 = 100$. Right: $+21 \times -21 = -441$. Answer: $10000 - 441 = \mathbf{9559}$. (Equivalently: $(100+21)(100-21) = 100^2 - 21^2 = 10000 - 441$.) Notice you've just used the *difference of squares* identity. Connect to algebra.
- **Tier down:** Stay with 102×99 , 103×98 (small differences). Build up the sign care before tackling larger gaps.

Teacher notes

- The mixed case is where most students give up and revert to long multiplication. **Don't let them.** The sign care is genuinely tricky for one day; by Day 6 it'll be a non-issue.
- The “tier up” connection to difference-of-squares (a^2-b^2 identity) is a real bridge to algebra. If a student spots it, celebrate.
- Some students will resist the negative right part. Show them the “borrow from left” form (above) as an alternative — both forms give the same answer.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Samskritam WA) — Nikhilam multiplication, mixed-sign examples.